Robot Inspection - Field Network Setup Guide

Spot (GXP) + Beryl AX robot LAN + Trimble X7 - Jetson Orin Nano, fully wireless operator laptop

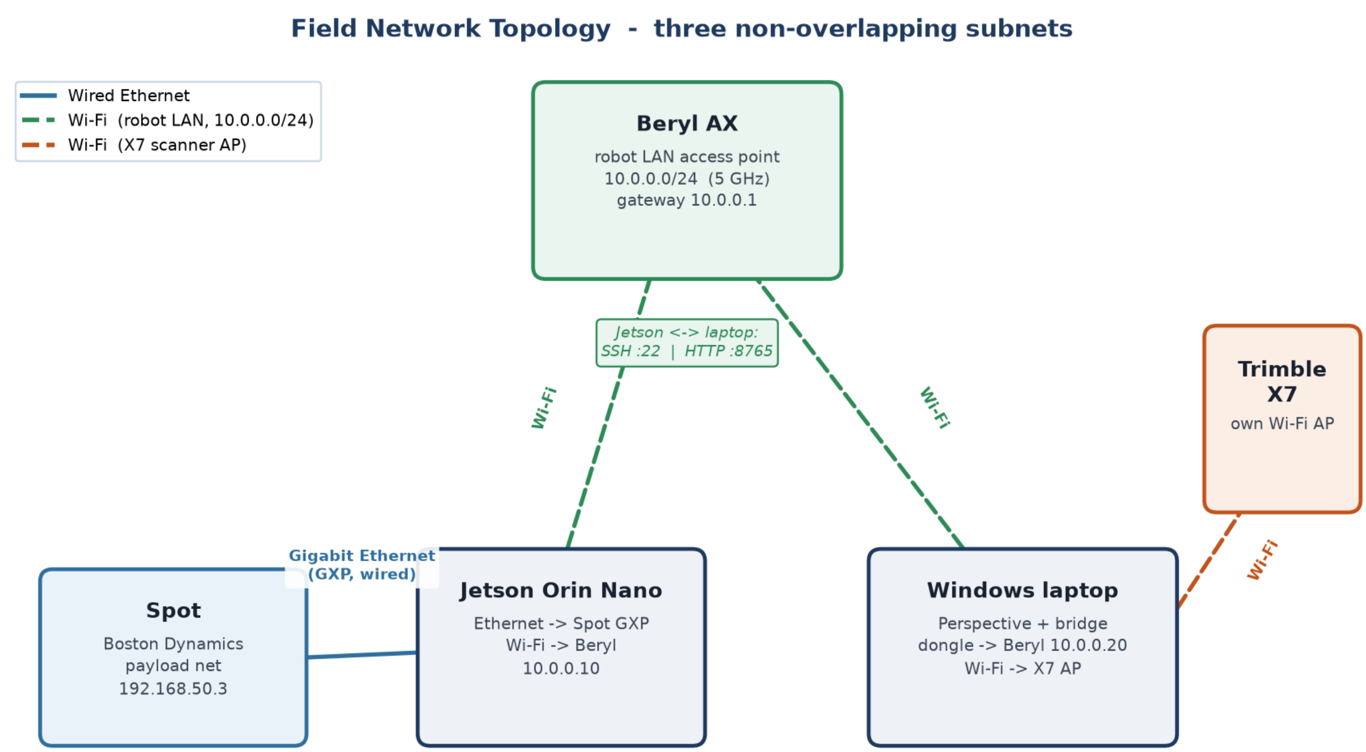
1. Overview

The system spans two computers, each owning a device only it can talk to. The Jetson (a Spot payload) runs ROS/autonomy and owns Spot, the Oak-D, and navX. The Windows laptop runs Trimble Perspective, the only software that can drive the X7, over the X7's own Wi-Fi. The two computers must message each other, so they share one network - the Beryl robot LAN. Everything is split across three subnets that must never overlap:

Subnet	Range	Carries	Link
Spot payload (GXP)	192.168.50.x	Jetson <-> Spot SDK (motion; localization is the Oak-D)	Wired Ethernet
Beryl robot LAN	10.0.0.0/24	Jetson <-> laptop (SSH :22, HTTP :8765)	Wi-Fi
Trimble X7	its own AP	laptop <-> X7 (Perspective)	Wi-Fi

Note: only the Jetson runs ROS, so ROS_DOMAIN_ID matters only there. The laptop is not a ROS node - it talks to the Jetson over plain SSH and HTTP.

2. Topology diagram



The Beryl is a hub the Jetson and laptop both join by Wi-Fi. Jetson<->Spot is a separate wired link that never touches the Beryl.

3. Hardware checklist

Item	Purpose	Notes / recommendation
Beryl AX (GL-MT3000)	Hosts the robot LAN (Wi-Fi AP)	Already owned
USB Wi-Fi adapter (laptop)	2nd radio: laptop -> Beryl while built-in Wi-Fi stays on X7	Netgear A6210 (MT7612U). Dual-band, external antenna, plug-and-play on Windows
Jetson Wi-Fi (M.2 E-key)	Jetson -> Beryl (single Ethernet is used by Spot)	Intel AX210 M.2 2230 if the slot is empty; or a USB Wi-Fi dongle / USB-Ethernet adapter

4. Beryl AX configuration

Admin UI at http://192.168.8.1 (factory default; join the GL-MT3000-xxx Wi-Fi, password on the bottom sticker). Configure as a standalone router/AP - no internet uplink is needed; ignore the 'No Internet Connection' banner and Exit the internet-mode wizard.

1. Mode: Router (default). Leave WAN unplugged - offline field LAN.
2. LAN subnet: 10.0.0.0/24, gateway 10.0.0.1 (moves the admin page to 10.0.0.1 - reconnect after applying).
3. Wi-Fi: SSID e.g. 'spot-field', WPA2/WPA3, strong password; prefer 5 GHz for the robot LAN.
4. DHCP static leases: Jetson -> 10.0.0.10, laptop USB-dongle MAC -> 10.0.0.20.
5. AP/client isolation OFF - the Jetson and laptop must reach each other.

5. Jetson (Orin Nano)

The single Gigabit Ethernet port goes to Spot's GXP. The Beryl link comes off the Jetson's Wi-Fi.

1. Wire the Jetson's Ethernet port to Spot's GXP / payload port (Spot at 192.168.50.3 by default - confirm yours).
2. Check for Wi-Fi: 'nmcli device status' and look for wlan0.
3. If wlan0 exists: join the Beryl SSID; reserve 10.0.0.10.
4. If not: install an Intel AX210 in the M.2 E-key slot (connect both antennas), or use a USB Wi-Fi dongle / USB-Ethernet adapter.
5. Confirm both links: ping 192.168.50.3 (Spot) and ping 10.0.0.20 (laptop).

field.env values on the Jetson (config/field.env):

```
SPOT_IP=192.168.50.3
TRIMBLE_WINDOWS_URL=http://10.0.0.20:8765
ROS_DOMAIN_ID=42
```

6. Windows laptop (two wireless NICs)

1. Built-in Wi-Fi: join the X7's own AP exactly as Perspective uses it today.
2. USB dongle: join the Beryl SSID; it should receive 10.0.0.20.
3. Windows routes by subnet automatically; neither interface needs to be a default gateway on an offline LAN.
4. Open the firewall so the Jetson can poll the bridge (run as admin):

```
New-NetFirewallRule -DisplayName "Trimble Bridge" -Direction Inbound -LocalPort 8765 -Protocol TCP
-Action Allow
```

- Bridge app Settings: jetson_host = 10.0.0.10, listen_port = 8765, windows_advertise_host = 10.0.0.20.

7. How the laptop and Jetson talk

- Laptop -> Jetson (SSH :22): 'Start Mission' SSHes in to build and launch the stack; 'Stop' SSHes a shutdown and scp's the digital twin back.
- Jetson -> laptop (HTTP :8765): the trimble_windows_bridge node POSTs scan requests / status and GETs /status every ~2 s. Pressing 'Next scan location' on the laptop bumps a counter the Jetson sees on its next poll, which releases the robot to the next vantage.

8. Verification (bench test before the field)

On the Jetson:

```
ping 10.0.0.20 && curl http://10.0.0.20:8765/status
ping 192.168.50.3
./scripts/field_preflight.sh mission
```

On the laptop:

```
ssh <user>@10.0.0.10 "echo ok"
```

Then a motion-safe end-to-end dry run (proposes goals, does not move Spot):

```
ROBOT_GOAL_BRIDGE=false ./scripts/run_field.sh mission
```

9. Troubleshooting

Symptom	Likely cause	Fix
Jetson can't curl :8765	Windows firewall / wrong dongle IP	Add the firewall rule; confirm dongle = 10.0.0.20
Laptop SSH hangs	No key / Jetson not on LAN	Set up an SSH key; confirm Jetson = 10.0.0.10
Perspective loses the X7	Built-in Wi-Fi left the X7 AP	Keep built-in Wi-Fi on the X7; robot LAN on the dongle
Routing conflicts	Overlapping subnets	Keep X7 AP, Beryl (10.0.0.x), Spot (192.168.50.x) distinct
Jetson has no wlan0	M.2 E-key slot empty	Install Intel AX210, or USB Wi-Fi / USB-Ethernet adapter